MRINAL PANDEY

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PROFESSIONAL SUMMARY

PhD-trained biological engineer with expertise in advanced 3D human-relevant cellular models, microfluidic platform development, and quantitative phenotyping for translational research. Proficient in mammalian cell culture, biochemical and molecular techniques (ELISA, Western blot, qPCR, immunofluorescence), and confocal microscopy. Designed and validated collagen-embedded spheroid systems to investigate tumor mechanics, invasion, and immune interactions by integrating imaging, transcriptomic, and protein-level analysis. Demonstrated ability to combine molecular and imaging datasets to identify mechanistic regulators, including STAT1/IFN pathway modulation under defined microenvironmental perturbations.

RESEARCH INTERESTS

- Precision Oncology
- Translational models for immuno-oncology drug discovery
- *In Vitro* Assay Development
- Multi-organ and organ-on-chip platforms
- Integration of imaging and transcriptomics for mechanism-of-action studies

PROFESSIONAL EXPERIENCE

Postdoctoral Associate *Cornell University, USA* February 2026 – Aug 2026

- Lead the design and implementation of a 3D tumor–NK cell interaction assay integrating collagen-embedded spheroids with controlled mechanical compression
- Provide technical mentorship to PhD students on microfluidic platform design, assay validation, and quantitative image analysis

Research Assistant *Cornell University, USA* January 2020 – Dec 2025

- Established robust 3D breast tumor spheroid culture systems, including collagen-embedded models and immune co-cultures
- Executed immunofluorescence staining workflows (fixation, permeabilization, antibody optimization, controls) and quantified target protein intensity
- Performed qPCR to validate RNA-seq transcriptomic findings, providing orthogonal confirmation of STAT1/IFN pathway suppression at the mRNA level
- Led end-to-end experimental design, data acquisition, and multi-modal analysis for mechanistic studies of tumor–microenvironment interactions

Teaching Assistant *Cornell University, USA* January 2020 – Jan 2025

- Designed, conducted and graded laboratory sessions for 100+ undergraduate students for Bioinstrumentation course
- Earned consistently positive feedback for clear communication, well-structured instruction, and approachable demeanor.

Master's Thesis Researcher *SciLifeLab, Sweden* September 2018 - Oct 2019

- Designed and developed a microfluidic device for ELISA based protein detection
- Developed protocols and validated immunoassays for host cell protein detection.

CHUST-KTH Intern *Brown University, USA* June 2018 - Aug2018

- Enhanced resolution of the PerkinElmer LabChip GX II system, achieving a twofold improvement in fragment size discrimination
- Advanced application of cfDNA size-based separation to support non-invasive prenatal diagnostics.

Quality Assurance Engineer *Infosys Limited, India* October 2014 - June 2017

- Contributed to Quality Assurance projects for Molina Healthcare (USA), collaborating with onshore teams through daily stand-up meetings to address client requirements and ensure timely implementation

- Coordinated with team leaders and junior trainees to streamline testing processes, improve deliverable quality, and align project outcomes with client expectations.

SELECTED RESEARCH EXPERIENCE

3D mechanical compression suppresses STAT1/IFN signaling in breast tumor spheroids (In review)

- Developed a three-layer microfluidic device using soft lithography
- Investigated tumor motility parameters using time-lapse BF and GFP images
- Bulk RNA sequencing followed by pathway analysis using Bioinformatics tools, Ingenuity Pathway Analysis (IPA) and Gene Set Enrichment Analysis (GSEA), that showed remarkable suppression of IFN signaling in compressed spheroids
- Immunofluorescence imaging to validate the suppression of STAT1 (identified molecule in IFN signaling) and its downstream regulators upon compression

Viscoelastic properties of tumor spheroids revealed by a microfluidic compression device and a modified power law model (RSC- Soft Matter, Paper)

- Applied dynamic compression using microfluidic device to identify the viscoelastic properties of breast tumor spheroids of different malignancy levels
- Quantified the relaxation dynamics and fitted the relaxation to a modified power-law model
- Using model fit, viscoelastic parameters, half relaxation and effective viscosity, were calculated

Mechanical compression regulates tumor spheroid invasion into a 3D collagen matrix (IOP Physical Biology Paper)

- Developed an in-vitro compression platform to compress breast tumor spheroids using static load
- Performed time-lapse imaging of spheroids under compression and recorded motility parameters of cells within the spheroids
- Quantified the invasion metrics of spheroids under compression

TECHNICAL EXPERTISE

3D Human-Relevant Models

- 3D spheroid and collagen-embedded systems
- Immune co-culture (NK–spheroids)

Biochemical & Molecular Techniques

- ELISA (sandwich immunoassay, host cell protein detection)
- Western blot (protein expression analysis), qPCR (transcriptomic validation)
- Immunofluorescence (target protein quantification)

Imaging & Quantitative Phenotyping

- Confocal microscopy, Time-lapse imaging, Image segmentation and feature extraction (Python/ImageJ)

Molecular & Systems Analysis

- Bulk RNA-seq analysis, Pathway enrichment (IPA, GSEA), Quantitative analysis using GraphPad Prism

Platform Development

- Soft lithography, PDMS microfabrication

Good Clinical Practice (GCP) - NIDA - Certified, Sep 2025 - Sep 2028

PUBLICATIONS

- **Pandey, M.**, Roach, K., Suh, Y. J., Segall, J. E., & Wu, M. (2026) 3D Mechanical Compression Suppresses STAT1/IFN Signaling in Breast Tumor Spheroids **(In review)**
- Muriuki, F., Roach, K., Suh, Y.J., **Pandey, M.,** Cheung, B.C., Segall, J.E. and Wu, M., (2026). A Microfluidic Platform for Studying Roles of Mechanical Compression in Tumor-Immune Cell Interactions in a 3D Extracellular Matrix. Journal of Visualized Experiments **(JoVE)**
- **Pandey, M.,** Zhu, B., Roach, K., Suh, Y. J., Segall, J. E., Hui, C. Y., & Wu, M. (2026). Viscoelastic properties of tumor spheroids revealed by a microfluidic compression device and a modified power law model. **(RSC, Soft Matter)**

- Suh, Y. J., Liu, M., Zhu, B., **Pandey, M.**, Cheung, B. C., Kim, J., ... & Wu, M. (2025). A microfluidic rheometer for tumor mechanics and invasion studies. **(Lab on a Chip)**
- **Pandey, M.,** Suh, Y. J., Kim, M., Davis, H. J., Segall, J. E., & Wu, M. (2024). Mechanical compression regulates tumor spheroid invasion into a 3D collagen matrix. **(Physical Biology),** 21(3), 036003.
- Zhang, T., Ameen, S., Ghosh, S., Kim, K., **Pandey, M.**, Cheung, B., ... & Schwarz, J. (2024). Enhanced extracellular matrix remodeling due to embedded spheroid fluidization. **(New Journal of Physics).**
- Suh, Y. J., **Pandey, M.,** Segall, J. E., & Wu, M. (2022). Tumor spheroid invasion in epidermal growth factor gradients revealed by a 3D microfluidic device. **(Physical Biology),** 19(3), 036002.

PRESENTATIONS

- [Talk] **Cornell University, Mechanobiology Symposium** (2026, May).
- [Talk] **APS March meeting** (2026, March).
- [Talk]**APS-DSOFT virtual meeting** (2025, Nov).
- [Poster] **GRC Physical Science of Cancer** (2025, March).
- [Poster] **BMES Annual meeting** (2025, October).
- [Talk] **APS March Meeting** (2022, March).

EDUCATION

PhD, Biological Engineering
Cornell University Honor society – Alpha Epsilon **January 2020 - December 2025**
MS, Biomedical Engineering
KTH, Royal Institute of Technology **August 2017 - October 2019**
BTech, Biochemical Engineering
Uttarakhand Technical University, India **August 2010 - June 2014**

LEADERSHIP AND VOLUNTEER

WE Lead Fellow *Cornell University* February 2023 - May 2023

- Women in Engineering Leadership Program
- Participated in a selective leadership program designed to advance women leaders in engineering and technical disciplines. Developed skills in engineering and leadership with daily workshops
- Trained in strategic decision-making, negotiation, and leading complex technical teams.

COLMAN Inclusive Leadership Program *Cornell University* June 2024 - June 2024

- Participated in a workshop designed to develop belonging, equip emotional intelligence and cultural awareness in future leaders
- Trained in strategic conflict-management, thoughtful culturally aware inclusive leadership.
- Volunteered W.O.M.E.N outreach event – Apr 2025

International Student Ambassador *KTH Sweden* Aug 2017 - June 2019

- Represented the KTH Sweden Medical Biotechnology program globally.
- Provided mentorship to prospective students on academics, research, and life in Sweden.

AWARDS AND HONOURS

- Ag and Bio Engineering Excellence Fund, CALS Cornell University (2025).
- Awarded travel grant from Cornell University, December 2024
- Received scholarship for CHUST-KTH Sweden - summer exchange to United States (2018)
- Received Government of India scholarship for academic excellence (2010-2014)
- Manuscript reviewer for the IOP journals (single-blind review), 2025-2026.